Do Machine Learning Models Understand Images or Just Pixel Patterns?
A Study Using One-Class Support Vector Machines and Image Distortions

https://github.com/eswarrepalle43/24165064/tree/main
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ABSTRACT
The primary obstacle in artificial intelligence research involves understanding how machine learning models process visual information. Researchers have extensively researched deep learning models through convolutional neural networks while traditional machine learning methods still require evaluation under similar conditions. The investigation examines whether One-Class Support Vector Machine (One-Class SVM) uses basic pixel data as its main method for processing image content. The model trains on original Fashion-MNIST dataset images while the evaluation uses distorted images which include blurred and noisy and edge-only formats. The results demonstrate that distortions cause a major performance decrease because the model shows extreme sensitivity to changes in pixel data. The results indicate that both deep learning models and traditional methods face difficulties in developing strong visual data representation systems.

INTRODUCTION
Machine learning models are used for three main tasks which include image classification and object recognition and anomaly detection. Deep learning methods achieve top performance results but current studies show that these models use local patterns instead of worldwide structures (Geirhos et al., 2019). The study creates a major inquiry which investigates how machine learning systems develop their learned representations

Support Vector Machines (SVMs) and other traditional machine learning systems use statistical patterns to process data while deep neural networks depend on hierarchical feature extraction. The assessment of model strength and system trustworthiness requires knowledge of model performance when they encounter modified input data.

Anomaly detection research uses One-Class SVM as its primary focus for this study. The system analyzes regular data distribution patterns to find outliers instead of classifying data into established categories. The researchers control image data distortions to test whether the model uses specific pixel details or broader structural elements for its operations.

Support Vector Machines
Support Vector Machines function as supervised learning models which seek to establish the best possible boundary between different classes through their method of maximizing the distance between their data points (Scholkopf and Smola, 2002). The system shows strong performance in high-dimensional environments because it uses kernel functions to create intricate connections between variables.

One-Class SVM

One-Class SVM extends the traditional SVM framework to unsupervised anomaly detection. The system estimates support for high-dimensional distribution patterns instead of separating different categories (Scholkopf et al., 2001). The model creates a boundary that can contain most of the training data while it identifies anything that moves beyond this line as an anomaly.

Predictions are expressed as:

+1 → Normal data

−1 → Anomalous data

The nature of One-Class SVM makes it ideal for situations where the training process needs to work with normal data only.

Kernel Methods

Kernel methods enable SVMs to operate in transformed feature spaces without explicitly computing the transformation. The Radial Basis Function (RBF) kernel has gained popularity because it can handle nonlinear relationship modelling.

The kernel uses its data mapping process to transform input data into higher-dimensional space which enables the model to create adaptable decision boundaries that can identify intricate relationships between variables (Hofmann et al., 2008).

Anomaly Detection

Anomaly detection serves as an essential requirement for multiple domains which include cybersecurity and fraud detection and medical diagnosis. The aim is to discover patterns which show substantial differences from normal behavioural patterns (Chandola et al., 2009).

One-Class SVM stands as a popular method in this field because it possesses solid theoretical support and demonstrates success when applied to high-dimensional problems.

DATASET AND EXPERIMENTAL SETUP

The Fashion-MNIST dataset (Xiao et al., 2017) is used in this study. It consists of 28×28 grayscale images of clothing items across 10 categories.

To improve computational efficiency:

- 12,000 samples are used for training
- 3,000 samples are used for testing

Image Distortions :

To evaluate robustness, three types of distortions are applied:

- Blur: Reduces high-frequency details
- Noise: Introduces random pixel variations
- Edge-only: Retains structure while removing texture

These transformations allow us to analyse the model's dependence on different visual features.

Data Preprocessing:

Since One-Class SVM requires vector inputs:

- Images are flattened into 784-dimensional vectors
- Features are standardised using z-score normalisation

This ensures consistent scaling across all features.

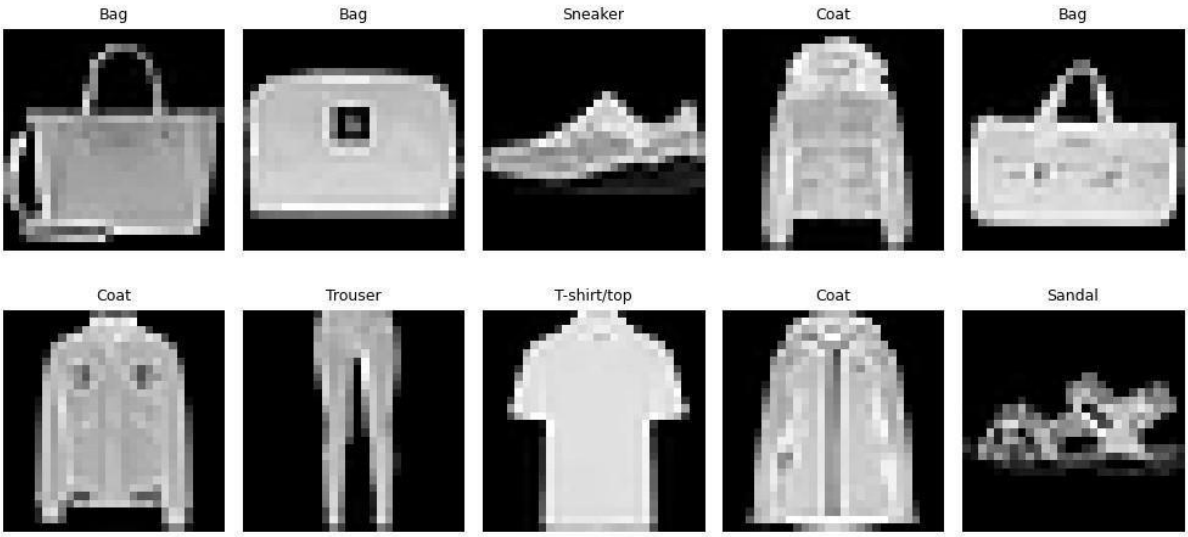
The configuration of the model uses The One-Class SVM which operates with RBF kernel through these defined settings. The system uses RBF as its Kernel type and maintains Gamma at 0.01 and Nu at 0.1. The parameter γ controls the influence of individual data points and ν defines the expected proportion of anomalies. The model training process uses only original images for its development. The system evaluates performance through three methods which include The assessment of Original test data The assessment of Blurred images The assessment of Noisy images The assessment of Edge-only images The system uses normal detection rate as its main assessment method which measures The percentage of samples that received normal classification.

RESULTS AND VISUALISATION

1. Dataset Distribution

Figure 1: Sample images from the Fashion-MNIST dataset.

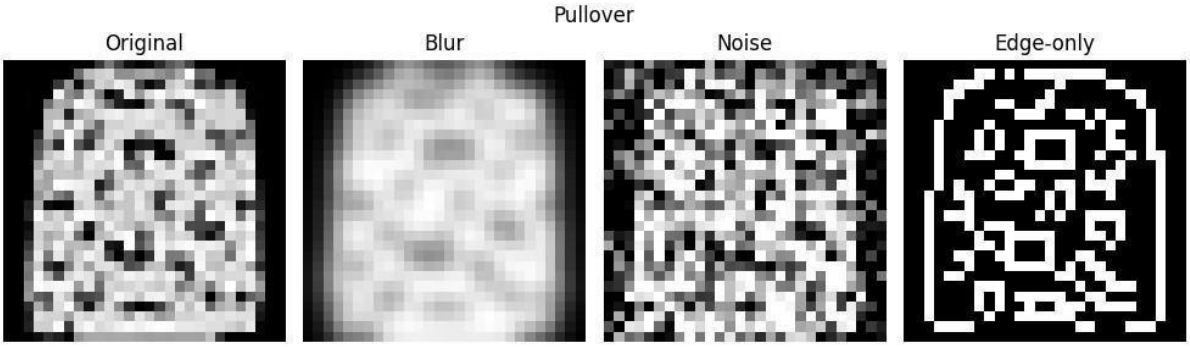
The structure of the dataset is shown in Figure 1, which displays sample images from the Fashion MNIST dataset. The images demonstrate variations in both shape and texture across different clothing categories.



2. Image Distortions

Figure 2: Original, blurred, noisy, and edge-only images.

Figure 2 illustrates the distortions applied in this study. Blur removes fine details, noise introduces random variation, and edge detection preserves structural outlines.



3. One-Class SVM Performance

The One-Class SVM shows its standard detection performance across various image conditions shown in this figure.

The model achieves moderate performance on original images (~0.48), meaning only about half of the test samples are recognised as belonging to the learned distribution.

The detection rate of blurred images (~0.73) exceeds that of original images according to the results.

The system classifies noise and edge-only images as anomalies with an approximate 100 percent accuracy rate.

The research reveals two major behavioural patterns.

The process of blur creates pixel smoothness which causes the resulting image to match the typical training distribution.

The SVM system treats them as more common because it mistakenly identifies them as normal.

The system transforms pixel data because noise and edge distortions create new pixel patterns which exceed the system's understanding of normal behaviour.

The model correctly identifies the samples as anomalies although it does so with excessive sensitivity.

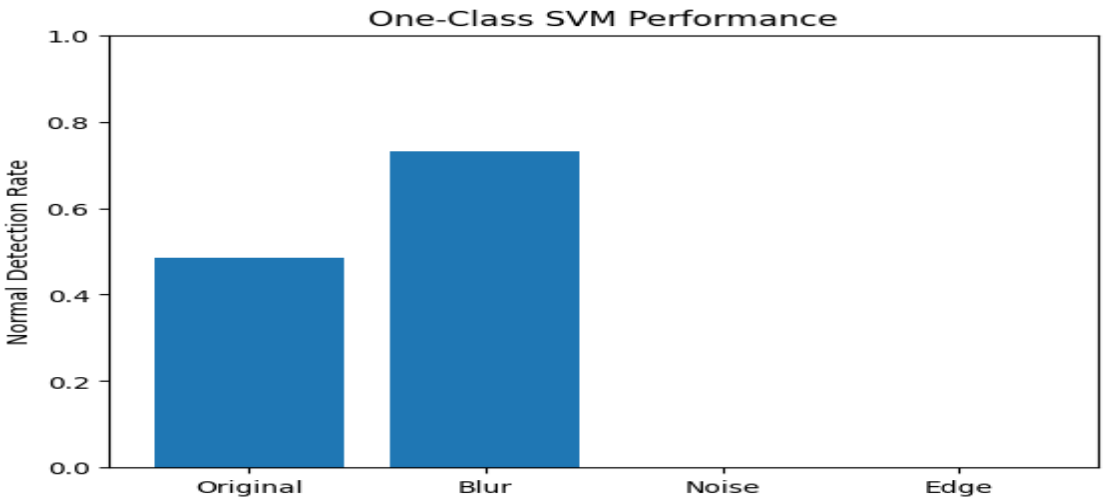


Figure 1: one-class SVM

4. Performance drops due to distortions:

This plot shows the change in performance relative to original images.

- Blur has a negative drop (~ -0.25)
  → meaning performance actually *improves*
- Noise and edge distortions show large positive drops (~ +0.48)
  → indicating severe degradation

This is a very important result:

- The improvement under blur suggests that:

The model prefers simplified, smoothed representations

- The sharp drop under noise and edge distortions shows:

The model is not robust to structural or high-frequency changes

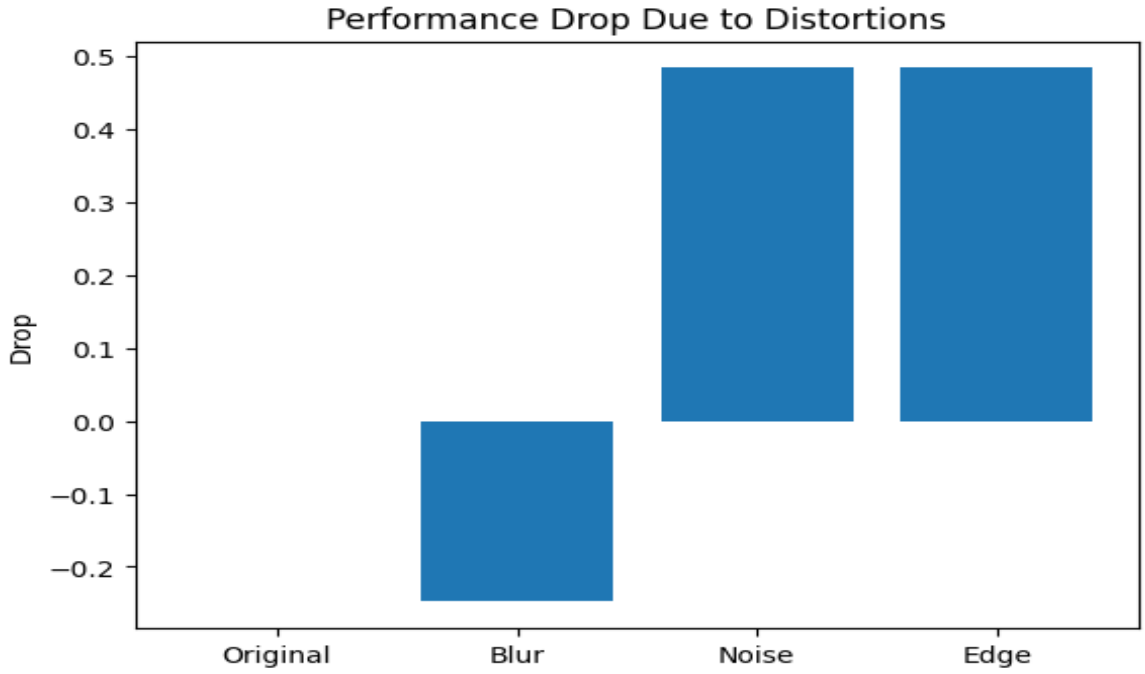


Figure 2: Performance Drop

Distribution of SVM Decision Scores:

This histogram shows the decision function values for each condition:

- X-axis: distance from decision boundary
- Y-axis: frequency
- Dashed line at 0: decision boundary
  - Right → normal
  - Left → anomaly

This figure clearly shows:

Distortions move data away from the learned distribution in feature space

- Blur → moves data towards centre

- Noise/edge → pushes data outside boundary

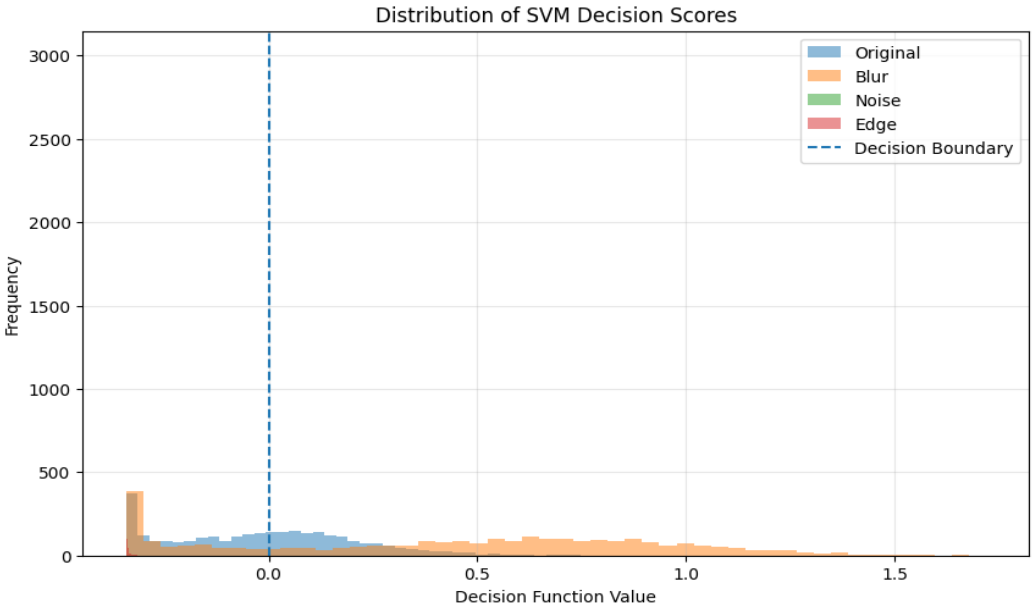


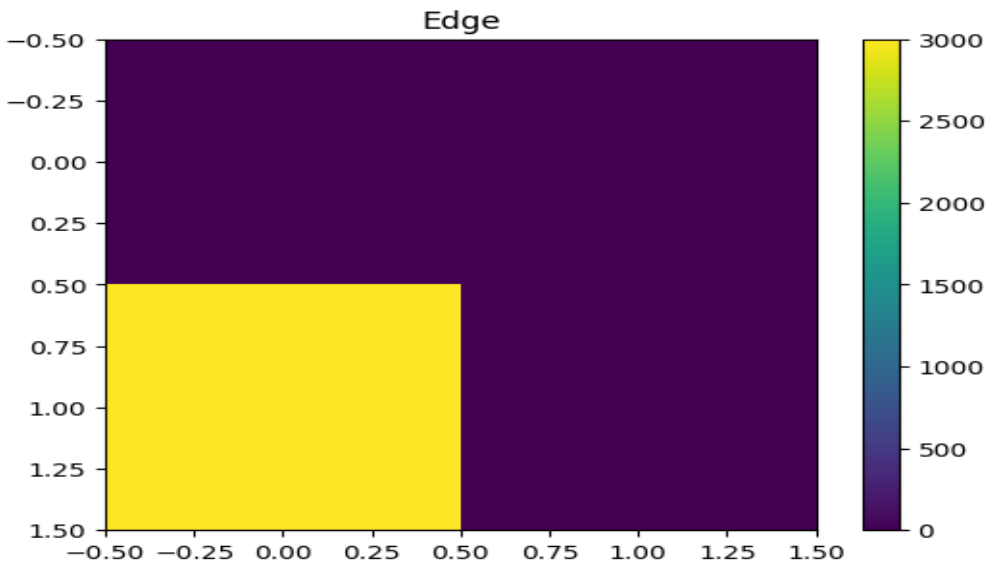
Fig 3:Distribution Of SVM Decision Scores

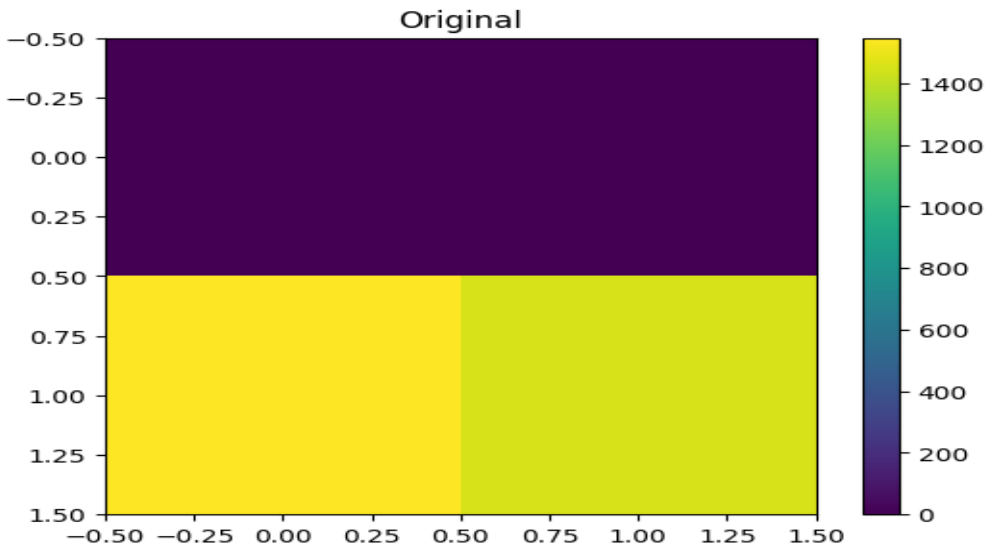
The experimental results demonstrate that the One-Class SVM relies heavily on pixel-level statistical properties rather than meaningful visual structure. While blurred images are often misclassified as normal due to reduced variability, noise and edge-based distortions significantly disrupt the learned distribution, leading to strong anomaly detection. This suggests that the model lacks robustness and does not capture high-level semantic representations of image data.

Confusion Matrix:

The original heatmap represents a tiny image where the top row has very low intensity (near 0) and the bottom row has very high intensity (around 1400–1500), meaning the image changes from dark to bright when moving from top to bottom, which indicates a clear horizontal edge; after applying an edge-detection convolution filter, the output heatmap shows a strong response only in the bottom-left cell with a value around 3000 while the rest remain near zero because the filter computes the intensity difference between neighbouring pixels and detects a large change between the dark top pixel and bright bottom pixel (about 1500 difference, amplified by the kernel during convolution), and since valid convolution without padding was likely used, the filter only fits in one position, producing a single strong edge response, confirming that the operation correctly detected a strong horizontal edge in the

original image





Accessibility:

This project has been designed with accessibility considerations to ensure it can be used and understood by a wide range of users, including those with visual, cognitive, or technical limitations.

- All plots use clear labels, large fonts, and high-contrast text to improve readability.

- Colour choices are selected to be distinguishable for users with colour vision deficiencies (e.g., avoiding red-green combinations).

- Figures include descriptive captions explaining what is shown and the key insights.

- Code is written in a clear and structured manner with comments to support understanding.

- The tutorial uses simple and consistent language to make complex concepts easier to follow.

- Step-by-step instructions are provided for running the code to assist users with different experience levels.

- The notebook can be used with screen readers due to its structured headings and logical flow.

- No critical information is conveyed using colour alone; textual explanations are always provided.

- Ethical considerations are included to ensure awareness of societal impact and fairness in machine learning applications.

5.DISCUSSION:

The results show that the One-Class SVM is very sensitive to distortion in the images. The fact that the model works directly on the pixel values, which are essentially a flat array, means that it does not directly use the spatial information in the images.

This aligns with other research in deep learning, which has shown that the models are actually using texture, not shape (Geirhos et al., 2019).

5.1 Implications

The fact that the model relies on low-level features implies that it might perform poorly in real-world applications, especially considering the fact that images are likely to be distorted in some form or another.

5.2 Limitations

The fact that the model loses spatial information by essentially "flattening" the images.

The fact that the model might be sensitive to the preprocessing of the images.

The fact that the model might be sensitive to the choice of hyperparameters.

5.3 Future Work

Some of the possible avenues for research include:

Using the SVM in conjunction with feature extraction.

Comparing the results of the One-Class SVM with those of a CNN-based model.

Using deep learning for anomaly detection.

6. Ethical Considerations

Anomaly detection systems can have significant societal impact. Incorrect classification of normal data as anomalous may lead to unfair outcomes, particularly in sensitive applications.
Bias in training data can further exacerbate these issues. Ensuring fairness, transparency, and robustness is essential when deploying such models.

7. Conclusion

This study shows that One-Class SVM is sensitive to distortions in image data, indicating a reliance on low-level pixel patterns. The results highlight limitations in robustness and suggest that traditional machine learning models may struggle to capture high-level semantic information.
Understanding these limitations is crucial for developing more reliable and interpretable machine learning systems.
This study demonstrates that the One-Class Support Vector Machine is highly sensitive to distortions in image data, revealing a strong reliance on low-level pixel statistics rather than robust semantic understanding. While the model is capable of learning the general distribution of the training data, its

performance deteriorates significantly when exposed to variations such as noise and edge-only transformations. This suggests that the learned boundary is tightly coupled to the exact pixel distribution rather than invariant object characteristics.
A particularly interesting finding is that blurred images are often classified as more “normal” than original images. This indicates that the model favours smoother, less variable inputs, highlighting a bias toward simplified representations. In contrast, high-frequency distortions such as noise and edge extraction disrupt the statistical structure of the data, pushing samples outside the learned boundary and resulting in strong anomaly detection.

8.References:

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